

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Section: I

System Name: Academic Elective Bidding & Allocation System

The Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable students to allocate 100 bidding points over their elective preferences and ensure the completion of prerequisite courses.	High	Pass: Sum of credit points is exactly 100 and student satisfies all the prerequisite course requirements. Fail: Credit points allocated exceed 100 and prerequisite courses are overlooked.	Facilitates proper allocation of credit points and avoids any unauthorized enrollment of courses.
FR-002	Functional	The system shall identify any timetable slot clashes that may occur between the elective options chosen by the	High	Pass: System identifies timetable clashes and prevents submission until resolved.	Avoids schedule conflicts and ensures a valid schedule for students.

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		student prior to bidding.		Fail: Conflicting course slots are submitted.	
FR-003	Functional	The system shall conduct an allocation algorithm that allocates students to seats using bids, credit weighting, and academic priority automatically.	High	Pass: The seats get allocated to the highest bidders only without exceeding section capacity. Fail: Over-enrollment is done or low bidders get seats ahead of high bidders.	Logical basis for core allocation that will automate seat allocation effectively.
FR-004	Functional	The system shall enable the Academic Registrar to set the courses offered, seats available, credit caps, and deadlines for bidding.	Medium	Pass: Changes in course configurations and deadlines show up immediately in the student portal. Fail: Changes do not take effect or bids continue to be accepted beyond the specified deadline.	Allows for administrative management of the course structure and schedule.
FR-005	Functional	The system shall produce and distribute individual	High	Pass: Students see allocated courses and can download	Concludes the cycle of bidding and gives official

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		reports of allocation and timetables to students after being approved by the registrar.		schedules without conflicts. Fail: Incomplete and/or unauthorized drafts are seen by students.	schedules to registered students.
NFR-001	Non-Functional (Performance)	The final elective allocation solver shall process up to 5,000 bids made by students in less than 30 seconds.	High	Pass: Log file benchmark shows that system executes a batch of 5,000 records in ≤ 30 s Fail: System takes more than 30 s to execute a batch or fails due to overload.	Facilitates high throughput batch processing during university-wide enrolment periods.
NFR-002	Non-Functional (Security & Integrity)	The system shall implement role-based access control (RBAC) and have an unalterable audit trail for all bids and administrative seat adjustments made.	High	Pass: Unauthorized users cannot access administrative interfaces; bid changes include a timestamp and user identifier. Fail: Changes made by unauthenticated users or lack of logs.	Ensures academic integrity and avoids any interference in students' selections or seating arrangements.